Enhancing the Mathematical Problem Solving Performance of Seventh Grade Students Using Schema-Based Instruction: Year 1, Design Experiment

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Abstract

In this design study, we developed and tested a curriculum that used schema-based intervention (SBI) in conjunction with self-monitoring (SM) instruction for teaching ratio and proportion word problems. Students were taught to self-monitor their problem solving skills using a four-step checklist. Eight intact sections of seventh graders were randomly assigned to either the intervention or control classes. Students in the intervention condition received SBI-SM, whereas those in the control group received instruction on the same topics using procedures outlined in the district-adopted mathematics textbook. Results indicated a significant treatment group effect (p <.001), favoring the SBI-SM, with regard to the amount of change between pretest and posttest. Although these findings support the use of SBI, the effect for transfer to novel and more complex problems was not evident. Implementation of SBI yielded important information about professional development, curriculum design, and instructional delivery.

Introduction

The Principles and Standards for School Mathematics issued by the National Council of Teachers of Mathematics (NCTM, 2000) emphasize the importance of problem based mathematics instruction. In school mathematics curricula, story problems that range from simple to complex problems represent "the most common form of problem solving" (Jonassen, 2003, p. 267). Problem solving provides the context for "learning new concepts and for practicing learned skills" (NRC, 2001, p. 421). Research with elementary and middle school children suggests that mathematical tasks involving story context problems are much more challenging than no-context problems (Cummins, Kintsch, Reusser, & Weimer, 1988; Mayer, Lewis, & Hegarty, 1992; Nathan, Long, & Alibali, 2002; Rittle-Johnson & McMullen, 2004). An approach to teaching problem solving that has shown to be effective emphasizes the role of the mathematical structure of problems. From schema theory, it appears that cognizance of the role of the mathematical structure (semantic structure) of a problem is critical to successful problem solution (Sweller, Chandler, Tierney, & Cooper, 1990). Schemas are domain or context specific knowledge structures that organize knowledge and help the learner categorize various problem types to determine the most appropriate actions needed to solve the problem (Chen, 1999; Sweller et al., 1990). For example, organizing problems on the basis of structural features (e.g., rate problem, compare problem) rather than surface features (i.e., the problem's cover story) can evoke the appropriate solution strategy. Ratio, proportion, and percent word problem solving was chosen as the content in our study, because proportionality is a challenging topic for many students (National Research Council, 2001) and current curricula typically do not focus on developing deep understanding of the mathematical problem structure and flexible solution strategies (NCES, 2003; NRC, 2001). In the current study, we investigated the effectiveness of SBI-SM instruction on students' ability to solve both familiar problems as well as novel and more complex problems (e.g., multistep, irrelevant information) when compared to a comparison group of students receiving conventional mathematics instruction. In addition, we evaluated the outcomes for students of varying levels of academic achievement.

Method

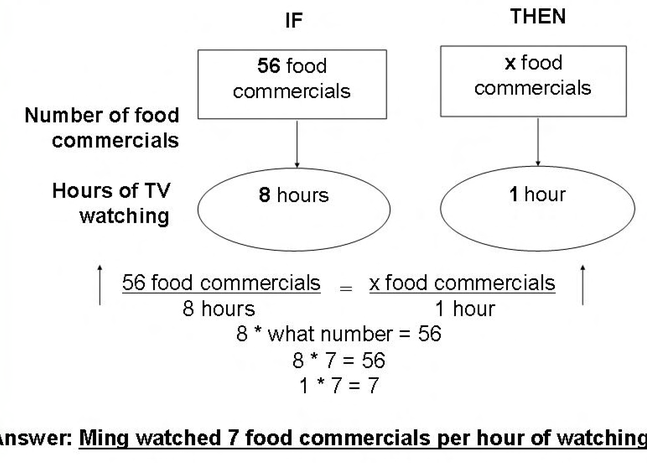
***Participants*:** One hundred fifty three (80 female) 7th graders and their teachers.

Procedure: Eight intact sections of seventh graders (n = 153) were randomly assigned to either the intervention (n = 74) or control condition (n = 79). Sections represented classrooms of students tracked on the basis of their mathematics performance: high ability (academic), average ability (applied), and low ability (essential). Both conditions were introduced to the same topics and received the same amount of instruction (i.e., 10 days).

Method (cont'd.)

The control group received instruction using procedures outlined in the district-adopted mathematics textbook. The SBI-SM condition used an instructional paradigm of teacher-mediated instruction followed by guided learning and independent practice in using schematic diagrams and SM checklists as they learned to apply the learned concepts and principles (see Figure 1 for sample materials). They also learned to use a variety of solution methods (cross multiplication, equivalent fractions, unit rate strategies) to solve word problems.

Proportion Problem: Ming watched TV for 8 hours on Saturday and saw 56 food commercials. How many food commercials did she watch each hour?



Answer: Ming watched 7 food commercials per hour of watching TV.

Figure 1. Sample intervention materials

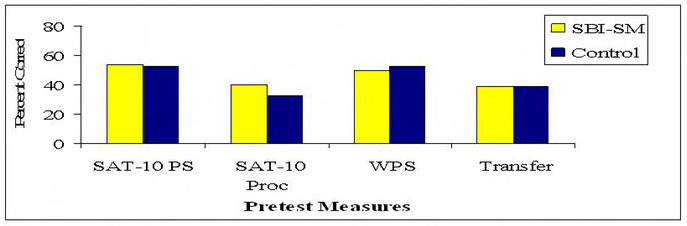
***Measures*:** Several measures were included to assess students' word problem solving performance and mathematics achievement. We developed a word problem solving (WPS) test and a transfer test using items from the TIMMS, NAEP, and state assessments. The WPS measure assessed ratio and proportion problem solving knowledge similar to the instructed content. The transfer test included novel and more complex items (e.g., multistep) (see sample items). Students completed the same tests at pretest and posttest. They also completed the problem solving and procedures subtests of the Stanford Achievement Test-10 at pretest.

Sample item from WPS test	*Sample item from Transfer test*
If there are 300 calories in 100 g of a certain food, how many calories are there in a 30 g portion of this food? A. 90 B. 100 C. 900 D. 1000 E. 9000	It takes 20 minutes per pound to cook a turkey. Mona's turkey weighs 7½ pounds. Peter's turkey weighs 9 pounds. How much longer will it take to cook Peter's turkey? A. 20 minutes B. 30 minutes C. 40 minutes D. 1½ hours

Results

The following research questions were analyzed using mixed effects models.

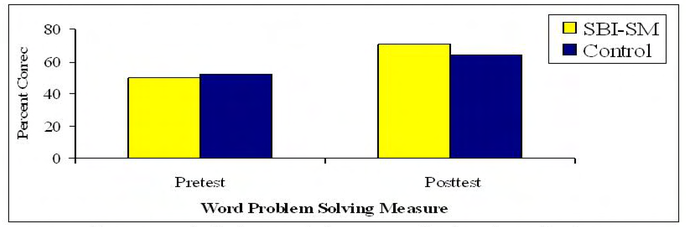
(1) Do the SBI-SM and control groups differ at pre-test on the SAT-10 Mathematics Tests and Problem-Solving measures?



Note. PS = Problem Solving; Proc = Procedures; WPS = Word Problem Solving measure.

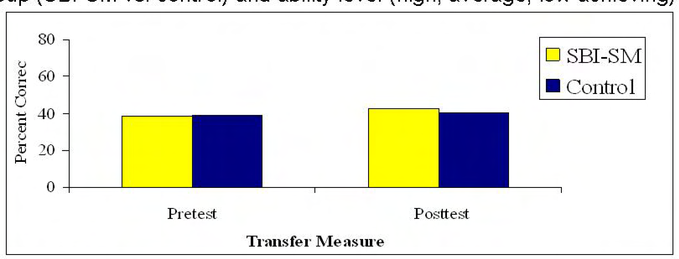
The groups did not differ at pretest on any measure.

(2) Do pretest-posttest differences on Word Problem-Solving vary by treatment group (SBI-SM vs. control) and ability level (high, average, low achieving)?



On average, the amount of change between pretest and posttest was significant (p <.001). A significant treatment group effect (p <.001) was found for pretest-posttest differences; however, the ability level effect was not significant (p =.07).

(3) Do pretest-posttest differences on the Transfer test differ by treatment group (SBI-SM vs. control) and ability level (high, average, low achieving)?



On average, the amount of change between pretest and posttest was not significant (p =.14). For pretest-posttest differences, neither treatment group effects (p =.28) nor ability level effects (p =.06) were found.

Conclusion

SBI-SM led to significant gains in problem-solving skills for students of varying ability levels. Developing deep understanding of the mathematical problem structure and fostering flexible solution strategies helped students in the SBI-SM group improve their problem solving performance. However, students' inability to transfer problem-solving knowledge may be explained, in part, by the short duration (i.e., 10 days) of the intervention, variations in the implementation fidelity, and the short duration of the professional development. Future research should address these issues based on the information the study yielded about professional development, curriculum design, and instructional delivery.